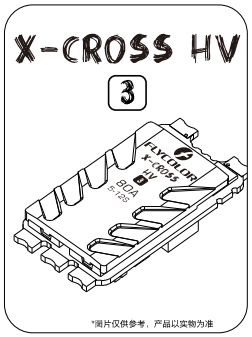
FLYCOLOR®

多旋翼飞行器
无刷电子调速器说明书

感谢您购买本产品！本产品功率强大，错误的使用可能导致人身伤害和设备损坏，强烈建议您在使用设备前仔细阅读本说明书并保存，严格遵守规定的操作程序。我们不承担因使用本产品或擅自对产品进行改造所引起的任何责任，包括但不限于对附带损失或间接损失的赔偿责任。我们有权在不经通知的情况下变更产品的设计、外观、性能及使用要求。



01主要特性

- ARM 32-bit Cortex 核心MCU，工作频率高达64MHz，较上一代MCU提升25%；
- 较上一代 ESC，优化了BLHeli-32 固件，超快响应速度，卓越的性能；
- 较上一代 ESC，优化设计，更佳的走线和元件布局，内置电流计；
- 高性能、低内阻Mosfet，不惧大电流；
- 宽电压支持，最高支持12S的锂电池，适用于X-Class级别无人机；
- 电调上电自动检测油门信号，支持普通PWM油门模式（1-2ms）的脉宽输入、Oneshot125（125-250us）、Oneshot42（41.7-83.3us）和 Multshot（5-25us）；支持所有Dshot 和Proshot数字信号；
- 尺寸小，重量轻，适合较窄机臂的无人机使用；
- 支持数字信号的数据回传，如温度、电压、电流、转速等.....；
- 信号线为双绞硅胶线，有效降低信号在铜线内传输所产生的干扰,使飞行更稳定；
- 铝合金散热器，帮助减缓温升。

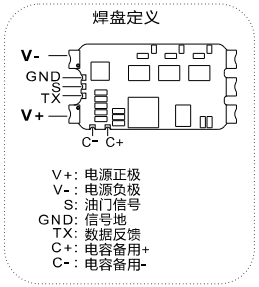
02 产品规格

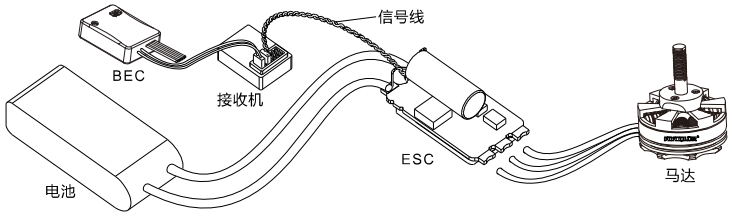
型号	持续电流 (散热良好)	瞬间电流 (散热良好)	BEC	锂电节数	重量 (供参考)	尺寸 (供参考)	LED设置	典型应用 (供参考)
X-Cross HV 3-60A	60A	80A	No	5-12S	25.5g	55x28x10mm	No	650-1000 X-Class
X-Cross HV 3-80A	80A	100A	No	5-12S	25.5g	55x28x10mm	No	650-1200 X-Class

*ESC BLHeli-32固件:**Flycolor_X_Cross_HV3_G071**;

03 连线示意图

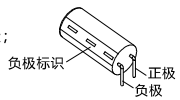
*为避免短路和漏电，请确保连接处绝缘良好。

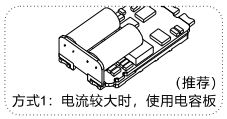




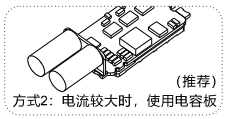
*图片仅供参考，产品以实物为准。

- 所有焊接要求良好的焊接技术，任何时候都需要避免因焊接而造成元件或线材之间短路；
- 电解电容引脚焊接越短越好，或者使用电容板避免大电流烧断引脚。焊接完电容建议打胶固定；
- 如电流较大时，建议将配件中的至少两个电容都焊接在ESC电源上、或者使用电容板；
- 无论任何时候都要注意极性，供电之前一定要反复检查；
- 根据个人安装方式，电解电容可以焊接位置不同，见下图。

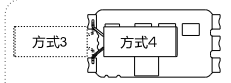




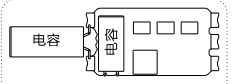
方式1：电流较大时，使用电容板



方式2：电流较大时，使用电容板



方式3/4：横向放置，朝内或朝外



方式5：电流较大时，放置2个电容

04 编程参数值

以下的参数需要通过BLHeliSuite32调参软件设置

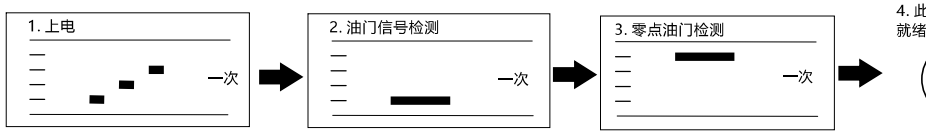
1. 启动功率（Rampup power）：
启动功率可以设置为从3%-150%的相对值。这是在启动和提高转速时允许最大功率。对于低转速，为了便于低反电动势电压检测，最大功率是被限制的。
启动功率也影响双向操作，参数是用来限制在更改转向时的功率。在启动过程中，实际功率取决于油门输入，可低于设定的最大启动功率，但最低是设定的四分之一。

2. 电机进角（Motor timing）：
电机进角可以设置为自动 或1°-31°，通常设置中等数值进角即适用于大部分电机，但如果电机运转不顺畅时,可以尝试改变进角。对于一些高感电机，其换向退磁时间较长，尤其在低速运转的时候，电机会在油门快速增加的情况下停转或者不顺畅。将进角改高会有有助于改善这个现象，因为高进角允许更长的换向退磁时间。

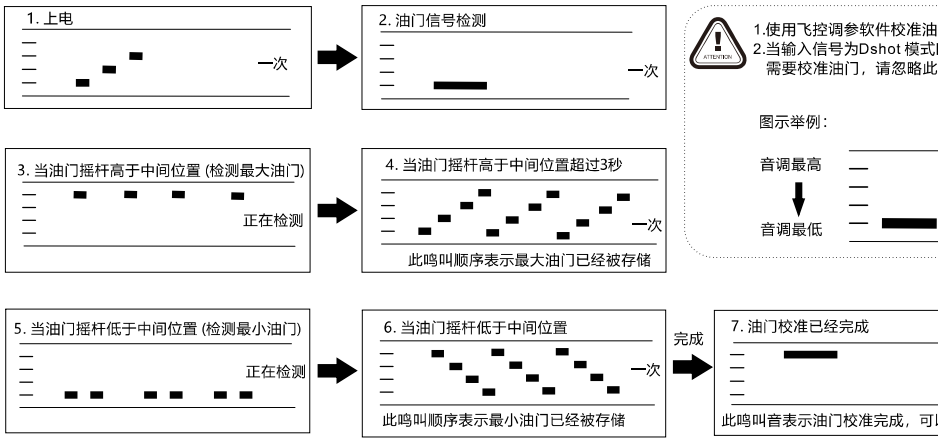
3.PWM频率（PWM frequency）：PWM Frequency Low - 低频率在低油门最需要的时候提供良好的主动制动；PWM Frequency High-高频率在更高的油门时使运行更平稳，或可设置为根据转速变化。更高PWM频率使电机运行更顺畅。频率可设定有可能会导致油门轻微移动时造成大的波动。

4. Demag补偿 (Demag compensation)：Demag 补偿是防止电机由于换向引起停转的一个功能，典型的现象是在快速增加油门时电机停转或不顺畅，尤行时。如前面所述，设置高进角可以帮助改善，但有可能降低效率。一般情况下，Demag补偿参数的值越高，保护越好。如果补偿值设置得太高，最大功
5. 正弦调制模式(Sine Modulation Mode)：正弦调制模式可以使运行效率提高百分之几，运行更平稳。注意：如果选择正弦调制模式，则可变pwm频
6. 最大加速度(Maximum Acceleration)：最大加速度可以设置在0.1% / ms-25.5% / ms之间，也可以设置为最大值，在这种情况下加速度不受限制。主要目的是避免加速不一致造成失步的情况。例如：当设置为10% / ms时，这意味着对电动机施加的功率不允许每毫秒增加10%以上。
7. 电机转向(Motor Direction)： 电机转向可以设置为正转/反转/双向3D/双向3D反转/双向soft/双向soft反转。 在双向模式下，油门中点为零点，中点以下为反转；当选择双向操作时，不可油门校准。
8. 启动音量(Startup Beep Volume)：设置通电期间的蜂鸣音音量。
9. 警报音音量(Beacon/Signal Volume)：设置警报音的音量。如果油门信号在零点位置超过一个设定的时间，电调将开始报警。高音量将会导致电机
10.警报音延迟(Beacon delay)：设置报警音开始之前的延时。
11.启用油门校准(Throttle Cal Enable)：如果禁用，将不能油门校准。
12. 最小油门，最大油门和中心油门(Min throttle, max throttle and center throttle)：设置电调的油门范围；中点油门只用于双向操作；设置值正
到 2000us的输入信号。对于其他输入信号，该值必须按比例设置。对于Dshot输入信号，这些设置无效。
13.温度保护(Temperature Protection)：可以启用或禁用。温度保护阈值可以设置，当温度高于阈值时,电机功率将降低；当温度高于阈值15℃,电机功
电机功率不会低于25%。
14.低转速功率保护(Low RPM power protect)：可以启用、禁用或自适应启用。禁用它可以以保证低KV电机在低电压运行时实现全功率，然而将增加
险并伴随着电机或ESC发热。
15.低压保护(Low Voltage Protection)：低压保护可以设置2.5V/节-4.0V/节锂电池，或者可以禁用。当启用时，如果电池电压低于设定阈值将限制电机
能主要用于固定翼飞机。
16.电流保护(Current Protection)：可以启用电流保护以限制电流。如果启用，则电流将限制在编程设定值的最大值。此为单个ESC的保护值 。
17.停车制动(Brake on stop)：可以启用或禁用。当设置启用时，在通电状态，油门在零点位置电机将会有拖刹，阻止电机转动。如果油门没有零点，
18.自动遥测(Auto Telemetry)：启用自动遥测时，ESC将以32毫秒的间隔自动输出遥测，无论是否有来自输入信号的遥测请求。
19.LED控制(LED Control)：可以控制发光二极管（如果ESC支持）。
20.堵转保护(Stall Protection)：可编程为Normal或Relaxed, Relaxed的堵转保护会增加ESC或电机损坏的风险，但当桨叶撞到障碍物时可以更快地恢
21.无阻尼模式(Non Damped Mode)：通过进行再生制动来实现阻尼模式，并实现固有的主动自由转动；由于制动而产生的损失会被主动自由转动减少
OFF-开启阻尼模式；ON-关闭阻尼模式。
22.S.BUS：如果选择了有效的S.BUS通道（0至16），则输入信号将被释放为SBUS。
23. S.PORT：如果选择了有效的S.PORT物理ID（1到28），则遥测格式将为S.PORT。请注意，只有使用USART1（端口PB6）进行遥测的ESC支持S.
BLHeliSuite32中显示“S.PORT Physical ID”编程参数，则表示ESC支持该项。
24.音乐设置(Music Note Config)：可以设置个性化音乐。

05 正常工作及提示音



06 油门校准



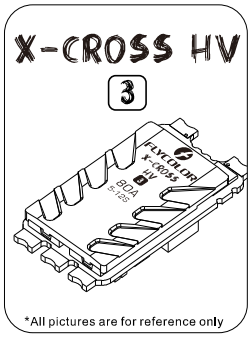
07 注意事项

- 电调接入飞行系统后,每次上电会自动检测输入的油门信号,然后执行相应的油门模式；
- 首次使用无刷电调或更换遥控设备后需要进行油门行程校准；Dshot 模式时，将不再需要校准油门；
- 请勿刷写除其它固件,以免损坏电调；
- 无论任何时候都要注意极性，供电之前一定要反复检查;
- 在插拔或者做任何连接时，请关闭电源;
- 当选择Dshot时，建议保留电调原本的双绞信号线中的地线，线连接正常;
- 当电机出现异常（如启动不顺畅）或者要求达到更高转速时进角参数;
- 请不要超出ESC工作电流、电压范围使用;
- 如需更多信息，请联系飞盈佳乐售后或者技术支持。

www.flycolor.net

User Manual
Multi-Rotor Brushless ESC

Thank you for using our product. Any Improper operation may cause personal injury or damage to the product and related equipments. This high power system for RC model can be dangerous ,we strongly recommend reading the user manual carefully and completely. We will not assume any responsibility for any losses caused by unauthorized modifications to our product. We have the right to change the design, appearance, performance and usage requirements of the product unannounced.



01 Main features

- ARM 32-bit Cortex MCU, frequency up to 64 MHZ, 25% higher than the previous generation of MCU.
- Compared with the previous generation of ESC, the ESC firmware is optimized, makes the throttle linearity smoother and the response faster.
- Compared with the previous generation of ESC, better routing and component layout, built in current meter.
- High performance and low ESR MOSFET , not afraid of large current.
- Wide voltage support, up to 12S liPo, suitable for x-class drone.
- Supports regular 1-2ms pulse width input, as well as Oneshot125 (125-250us), Oneshot42 (41.7-83.3us) and Multishot (5- 25us). The input signal is automatically detected by the ESC upon power up.
  All Dshot and Proshot signals are supported .
- Small size, light weight, suitable for the drone with narrow arms.
- Supports telemetry, and delivers the digital data such as temperature, voltage, current, speed, etc.
- The silicone twisted-pair of the throttle signal cable effectively reduces the crosstalk caused by signal transmission, and makes flight more stable.
- Aluminum heat sink effectively slow down the temperature rise.

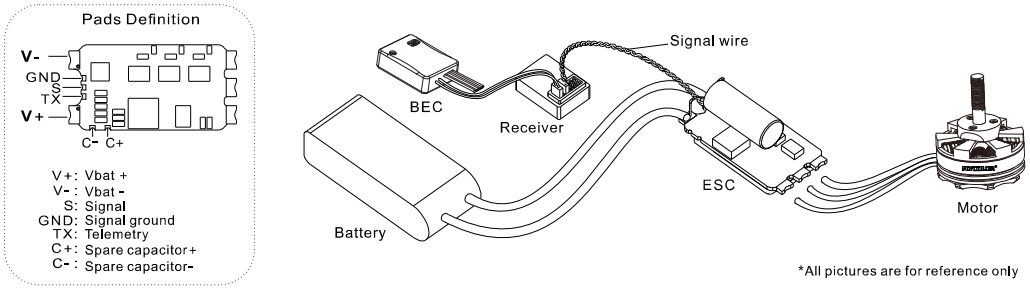
02 Specifications

Model	Con. Current (good heat dissipation)	Burst Current (good heat dissipation)	BEC	LiPo	Weight (For reference)	Size (For reference)	LED	Typical Applications (For reference)
X-Cross HV 3-60A	60A	80A	No	5-12S	25.5g	55x28x10mm	No	650-1000 X-Class
X-Cross HV 3-80A	80A	100A	No	5-12S	25.5g	55x28x10mm	No	650-1200 X-Class

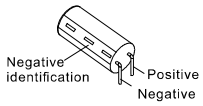
*ESC BLHeli-32 firmware: **Flycolor_X_Cross_HV3_G071**;

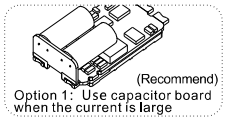
03 Connect diagram

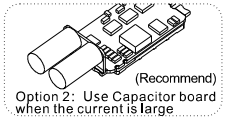
*Please ensure all solder joints are insulated with heat shrink where necessary.

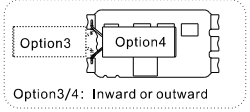


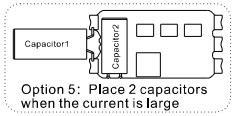
- All welding requires good welding technology, short circuit between the element or the wire should be avoided at any time.
- The shorter the welding of electrolytic capacitor pin, the better, to avoid burning off the pin with high current.
- If the current is large, it is recommended to weld two capacitors in the accessories on the ESC power supply. or use capacitor board. It is recommended to glue and fix the capacitor after welding.
- Observe polarity at all times. Check and double check before applying power.
- According to personal installation method, electrolytic capacitor can be welded at different positions, as shown in the figure below.









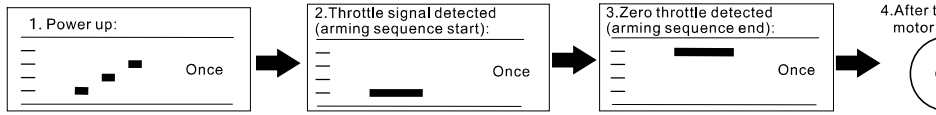


04 Programming parameter

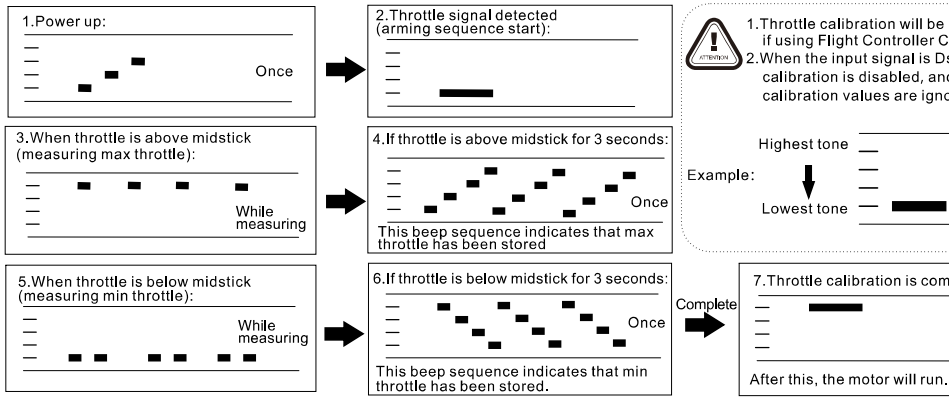
Programming parameters below can be accessed from the configuration software (BLHeliSuite32):

1. **Rampup power:** Rampup power can be set to relative values from 3% to 150%. This is the maximum power that is allowed when ramping up at low rpms and during startup. For low rpms, the maximum power to the motor is limited, in order to facilitate detection of low BEMF voltages. Rampup power also affects bidirectional operation, as the parameter is used to limit the power applied during direction reversal. During startup, the actual applied power depends on throttle input, and can be lower than the maximum level set by the rampup power parameter, but the minimum level is a quarter of the maximum level.
2. **Motor timing:** Motor timing can be set between approximately 1° and approximately 31° in approximately 1° increments (actual accurate values here are 15/16ths of a degree).Typically a medium setting will work fine, but if the motor stutters it can be beneficial to increase timing. Some motors with high inductance can have a very long commutation demagnetization time. This can result in motor stop or stutter upon quick throttle increase, particularly when running at a low rpm. Setting timing to high will allow more time for demagnetization, and often helps.
3. **PWM frequency:** Motor pwm frequency can be programmed in a range. Support variable pwm frequency where the pwm frequency is controlled by motor RPM. -Low frequency for low throttle gives good active braking where it is most needed. -High frequency for higher throttle makes running smoother.
4. **Demag compensation:** It is a feature to protect from motor stalls caused by long winding demagnetization time after commutation. The typical motor stop or stutter upon quick throttle increase, particularly when running at a low rpm. As mentioned above, setting high commutation timing no but at the cost of efficiency. Generally, a higher value of the it gives better protection. If it is set too high, maximum power can be somewhat reduce
5. **Sine Modulation Mode:** It gives a few percent more efficient running &smoother running. Note: if it is chosen, then variable pwm frequency is c
6. **Maximum Acceleration:** Maximum acceleration can be set between 0.1%/ms and 25.5%/ms. It can also be set to maximum, in which case acce limited. Limiting acceleration is primarily intended as a backup parameter that can be used in cases where too hard acceleration gives desyncs. W e.g. 10%/ms, it means that the power applied to the motor is not allowed to increase by more than 10% per millisecond.
7. **Motor Direction:** Motor direction can be set to fwd/rev/bidirectional 3D/bidirectional 3D rev/bidirectional soft and bidirectional soft rev. In bidire center throttle is zero and above is fwd rotation and below is reverse rotation. When bidirectional operation is selected, throttle calibration is disab
8. **Startup Beep Volume:** Sets the volume of beeps during power up.
9. **Beacon/Signal Volume:** Sets the volume of beeps when beeping beacon beeps. The ESC will start beeping beacon beeps if the throttle signal for a given time. Note that setting a high beacon strength can cause hot motors or ESCs!
10. **Beacon delay:** Beacon delay sets the delay before beacon beeping starts.
11. **Throttle Cal Enable:** If disabled, throttle calibration is disabled.
12. **Min throttle, max throttle and center throttle:** These settings set the throttle range of the ESC. Center throttle is only used for bidirectiona The values given for these settings are for a normal 1000us to 2000us input signal, and for the other input signals, the values must be scaled. For l signal, these settings have no effect.
13. **Temperature protection:** It can be enabled or disabled. And the temperature threshold can be programmed The programmable threshold is pr as a support for hardware manufacturers to use, as different hardwares can have different tolerances on the max temperatures of the various com
14. **Low RPM power protect:** Power limiting for low RPMs can be enabled, disabled or on adaptive. Disabling it can be necessary in order to achi on some low kV motors running on a low supply voltage. However, disabling it increases the risk of sync loss, with the possibility of toasting motor
15. **Low Voltage Protection:** Low voltage protection can be set between 2.5V and 4.0V per lipo cell. Or it can be disabled. When enabled, it will lim applied to the motor if the battery voltage drops below the programmed threshold. This feature is primarily intended for fixed wing crafts.
16. **Current Protection:** Current protection can be enabled to limit current. If enabled, then current will be limited to maximum the programmed val
17. **Brake on stop:** Brake on stop can be set between 1% and 100%, or disabled. When not disabled, brake will be applied when throttle is zero. For throttle, this setting has no effect.
18. **Auto Telemetry:** When it is enabled, the ESC will autonomously output telemetry at 32ms intervals, regardless of whether or not there are teler requests from the input signal.
19. **LED Control:** LEDs can be controlled on ESCs that support it.
20. **Stall protection:** It can be set to normal or relaxed. Relaxed increases the risk of damage to ESC or motor but can recover faster when props hi
21. **Non Damped Mode:** Damped light mode is implemented by doing regenerative braking, and inherently active freewheeling is also implemente losses due to braking are counteracted by the reduced losses of active freewheeling. **OFF-** Damped light is available ; **ON-** No Damped ligh
22. **S.BUS:** If a valid S.BUS channel (0 to 16) is selected, then the input signal will be interpreted as S.BUS.
23. **S.PORT:** If a valid S.PORT physical ID (1 to 28) is selected, then the telemetry format will be S.PORT. Note that only ESCs that use USART1 (p telemetry support S.PORT. If the "S.PORT Physical ID" programming parameter shows up in BLHeliSuite32, then your ESC supports it.
24. **Music Note Config:** Set up personalized music.

05 Beeps-Normal operation



06 Beeps - Throttle calibration



07 Attention

- ESC will automatically detect the input throttle signals every time as soon as it powered on, and then execute the corresponding signal-receiving mode.
- User need to calibrate the throttle range when starting to use a new ESC or another transmitter. When the input signal is Dshot, throttle calibration is disabled.
- Please don't flash any other firmware to avoid damage to the ESC.
- Observe polarity at all times. Check and double check before applying power.
- Power off before unplugging ,plugging in or making any c
- It is suggested that keep the ground wire in the original sig connecting well.
- When some abnormality occurs in ESC driving the motor motor to reach a higher RPM, user can try to change the ti
- Please do not exceed the current and voltage range.
- Please contact Flycolor sales or technical support for mor